Using Data to Verify Whether Ann Taylor Clothes Are True to Size for Women in Washington State

CSE 163 Final Project — Part 2: Exploratory Data Analysis | *Elly Lee* | *Student #2163622*

1. Research Questions

RQ1: How do the average body measurements (sleeve length, inseam, shoulder width, waist circumference) of women in Washington state compare to the U.S. national average?

RQ2: How do Washington state women's body measurements align with Ann Taylor's published size guide (Classic and Petite), and which product categories show the greatest sizing mismatch?

RQ3: Is Washington state's female demographic composition distinct enough from the U.S. average to warrant region-specific sizing considerations for retailers?

RQ4: Are the body measurement differences between Washington state women and the U.S. average statistically significant enough to justify region-specific sizing adjustments?

2. Motivation

The U.S. women's apparel industry is largely calibrated to the body proportions of white American women. Washington state has one of the highest concentrations of Asian-American residents of any U.S. state — a population whose women tend to have smaller and shorter body proportions on average. However, women of other ethnicities in Washington may also be more petite, and this project does not assume Asian-American women are the sole driver of any observed difference.

As a sales associate at the Ann Taylor location in Bellevue, WA — the only Ann Taylor store in the entire state — I have directly observed that **"too big/small"** is consistently the most frequently selected return reason in our store's return processing system. Because this single store serves all of Washington state, a sizing mismatch here affects Ann Taylor's entire WA customer base. Notably, Ann Taylor already carries a Petite fit category — yet returns citing size mismatch persist even on petite products, suggesting the existing petite line is still not adequately proportioned for Washington state women. This project aims to quantify that gap with data, and to provide a measurable foundation for more targeted sizing decisions in the Pacific Northwest market.

3. Datasets

Dataset	Source	Used For
ANSUR II (Women), 2012	openlab.psu.edu/ansur2	Structural measurements: sleeve length, inseam, shoulder width
NHANES 2015-2018 (Women)	openlab.psu.edu/nhanes	Circumference measurements: waist, height, weight (pre-merged CSV, women 20+, all in mm)

ACS PUMS 2020-2024 (WA + U.S.)	census.gov/programs-surveys/acs/microdata/access.html	Female racial/ethnic demographic weights for WA and national
Ann Taylor Size Guide (Classic + Petite)	anntaylor.com — via "Size Chart" button on any product page	Reference sizing standard; manually transcribed May 2026

4. Method

The EDA was implemented in two Python modules: **analysis.py** (helper functions for loading, cleaning, summarizing, and visualizing data) and **eda.py** (main pipeline calling those functions for each dataset). Key steps and deviations from the original proposal are described below.

Data loading: ANSUR II loaded directly from CSV with no filtering required. The Penn State NHANES CSV is already pre-merged — demographics and body measures are combined in one file, filtered to women aged 20+, so no SEQN join was needed (a deviation from the proposal). ACS PUMS national file arrived split into four parts (psam_pusa–d.csv), which were concatenated using pd.concat(). Both PUMS files were filtered to SEX == 2 (female).

Key discovery: ANSUR II has no race/ethnicity column: The proposal assumed ANSUR II could be broken down by race/ethnicity for demographic weighting. In practice, the dataset contains only a Branch column (military job category) with no race data. As a result, ANSUR II measurements are reported as national averages only, without demographic weighting. Only NHANES measurements (waist, height, weight) could be demographically weighted via PUMS.

Race/ethnicity harmonization: NHANES uses RIDRETH3 codes (1-7); PUMS uses RAC1P + HISP. Hispanic origin was given priority over RAC1P per Census Bureau convention, producing a unified race/ethnicity label used across both datasets.

Demographic weighting: PWGTP person weights computed proportional racial/ethnic composition of WA and national female populations from PUMS. These weights were applied to NHANES group means (grouped by race_label) via compute_weighted_means() to produce Washington-representative vs. national average body measurement estimates.

Ann Taylor size guide: Expanded from the proposal's single DataFrame to four: classic_tops, classic_bottoms, petite_tops, petite_bottoms. This was necessary after discovering that (1) bottoms have a separate Curvy fit subcategory with distinct waist/hip ranges, and (2) the Petite line extends to a larger maximum size for bottoms (18P) than for tops (16P), requiring separate DataFrames with different row counts. Classic tops and bottoms both max at size 18. All measurements converted from inches to cm (* 2.54).

5. EDA Results

5.1 ANSUR II Women Dataset

Size: 1,986 rows x 99 columns. Each row represents one U.S. Army servicewoman. Columns include 93 body measurements (all in mm), derived fields (BMI, BMI_class, Height_class), and demographic identifiers (Branch, Gender).

Missing data: No missing values detected across all 99 columns, verified via df.isnull().sum() — all columns returned 0. The dataset is fully complete.

Variables of interest: sleeveoutseam (sleeve/arm length, mm), crotchheight (inseam/leg length, mm), biacromialbreadth (shoulder width, mm), waistcircumference (waist, mm). These four measurements directly correspond to Ann Taylor's published size guide dimensions (sleeve, inseam, shoulder, waist) and are central to RQ1 and RQ2. Branch is the only available categorical variable; it does not relate to our research questions and is noted as a dataset limitation.

Variable	Mean	Std	Min	Q1	Median	Q3	Max
sleeveoutseam (mm)	543.6	29.4	410	525	542	563	662
crotchheight (mm)	782.3	44.6	610	753	780	812	947
biacromialbreadth (mm)	365.3	18.3	283	353	365	378	422
waistcircumference (mm)	860.9	99.9	611	790	852	925	1334

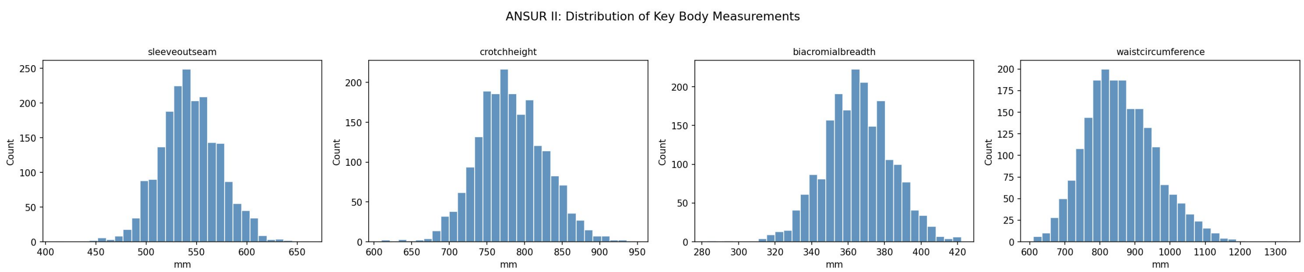

Figure 1. *Distribution of four key ANSUR II body measurements (mm). All four distributions are approximately normal. Waist circumference shows the widest spread (std = 99.9 mm), reflecting natural variation in body composition. These distributions confirm the t-test normality assumption for RQ4.*

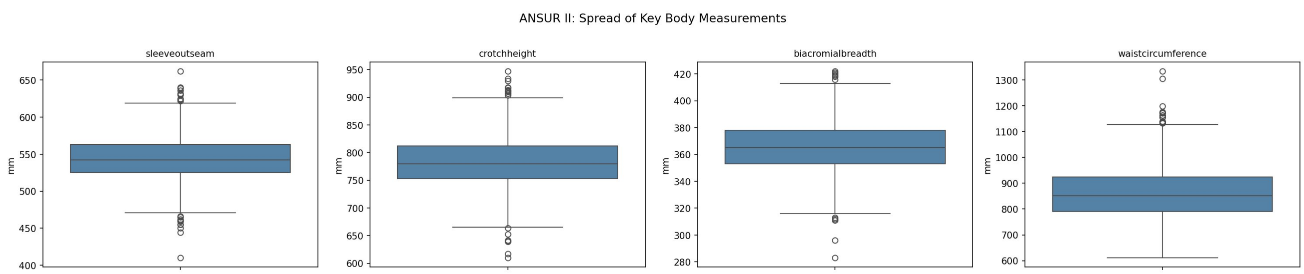

Figure 2. *Box plots of the same four ANSUR II measurements, revealing spread and outliers. Waistcircumference has the most outliers on the high end, while sleeveoutseam and crotchheight are tightly distributed around their medians. Box plots were chosen over a second histogram because they more clearly show quartile ranges and outlier positions.*

5.2 NHANES 2015-2018 Women Dataset

Size: 5,449 rows x 13 columns. Each row represents one civilian U.S. woman aged 20+. Columns include body measurements (BMXWAIST, BMXHT, BMXWT, BMXBMI), demographics (RIDRETH3, RIDAGEYR), pregnancy status (RIDEXPRG), and sampling weights (WTMEC2YR, WTMEC4YR, combinedWeight).

Missing data: 9,304 total missing values detected via df.isnull().sum(). However, the majority are from two columns irrelevant to this analysis: WTMEC4YR is 100% missing (5,449 values) — this column is unused since we use WTMEC2YR instead — and RIDEXPRG (pregnancy status) has 3,277 missing, expected since it applies only to women of reproductive age. For key measurement columns, BMXWAIST has 384 missing values (7.0%), and BMXHT and BMXWT each have 63 missing (1.2%). Rows with missing measurement values are dropped during analysis using dropna().

Variables of interest: BMXWAIST (waist circumference, mm) — connects to RQ1/RQ2 for sizing comparison; BMXHT (height, mm) and BMXWT (weight, kg) — used as additional body size indicators for RQ1; RIDRETH3 (race/ethnicity code, 1-7) — essential for demographic weighting in RQ1/RQ3; WTMEC2YR (2-year sampling weight) — must be used to produce nationally representative estimates per CDC guidelines.

RIDRETH3 (categorical): Non-Hispanic White: 1,763 **|** Mexican American: 1,523 **|** Non-Hispanic Black: 1,241 **|** Non-Hispanic Asian: 716 **|** Other: 206

Variable	Mean	Std	Min	Q1	Median	Q3	Max
BMXWAIST (mm)	989.1	172.6	579	862	974	1095	1716
BMXHT (mm)	1595.6	71.2	1297	1547	1595	1643	1893
BMXWT (kg)	77.1	21.8	32.4	61.3	73.0	88.1	219.6

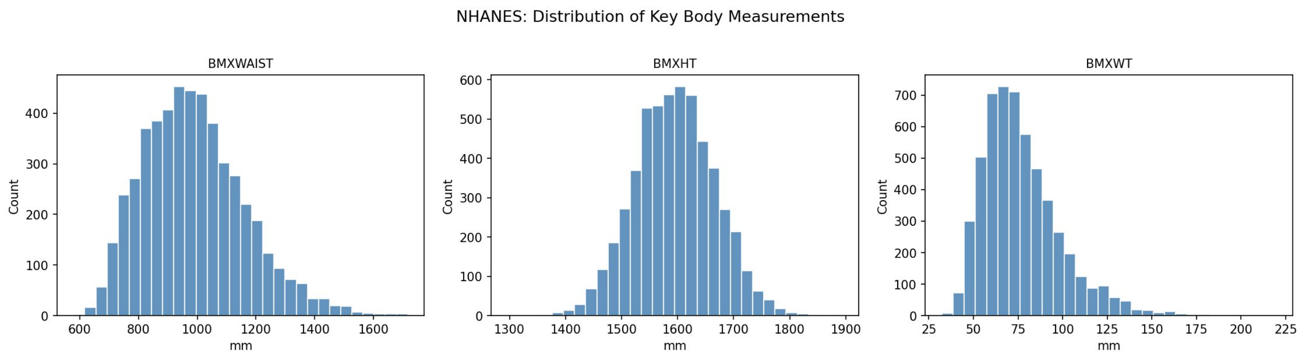

Figure 3. *Distribution of waist circumference, height, and weight in the NHANES Women dataset. Waist circumference is right-skewed, reflecting the known distribution in the U.S. population. Height is approximately normal. Weight shows a long right tail consistent with population-level obesity trends. The large sample size (n = 5,449) justifies t-test use despite the waist skew, per the Central Limit Theorem.*

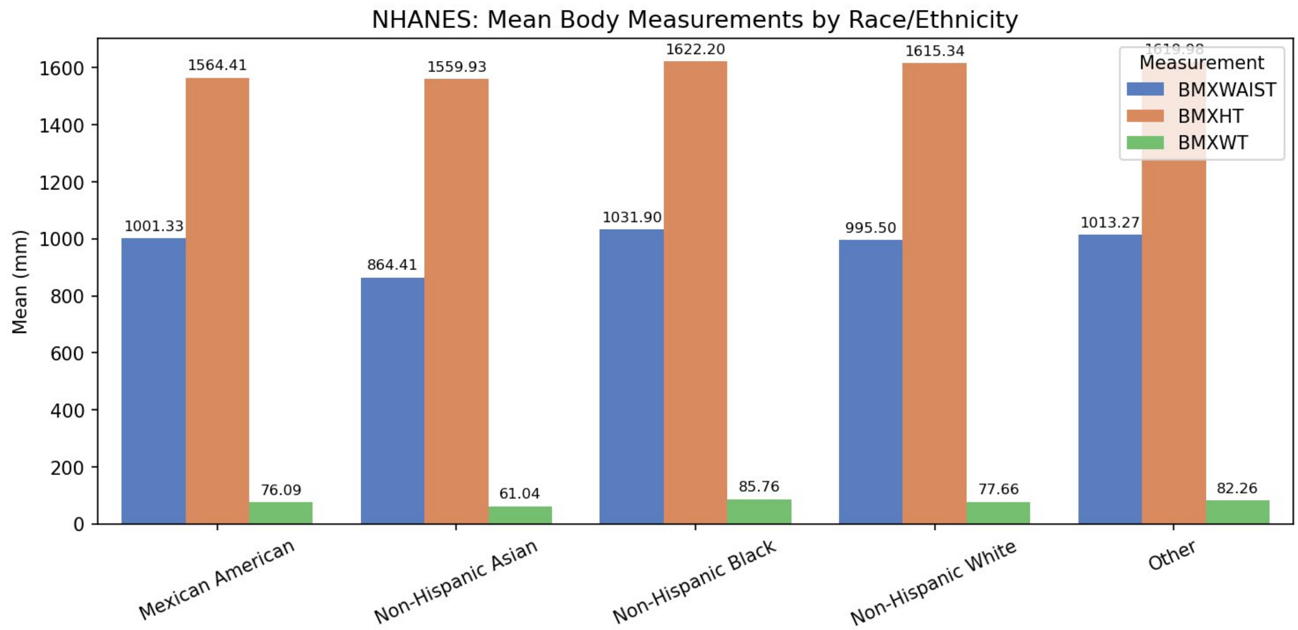

Figure 4. *Mean body measurements by race/ethnicity in NHANES. Non-Hispanic Asian women have the smallest mean waist (864.4 mm) and weight (61.0 kg) of all groups — notably lower than Non-Hispanic White (995.5 mm waist) and Non-Hispanic Black (1031.9 mm waist). This pattern directly motivates the hypothesis that Washington state's higher Asian population proportion contributes to a lower average waist measurement relative to the national norm.*

5.3 ACS PUMS 2020-2024 Dataset

Size: WA file (psam_p53.csv): 196,332 rows x 4 columns after filtering to female respondents (SEX == 2) and retaining only SEX, RAC1P, HISP, PWGTP. National file: 8,184,642 rows x 4 columns (concatenated from psam_pusa.csv through psam_pusd.csv). Each row represents one survey respondent.

Missing data: No missing values in SEX, RAC1P, HISP, or PWGTP in either the WA or national file, verified via df.isnull().sum() — all four columns returned 0 for both datasets.

Variables of interest: RAC1P (race, 1-9 codes), HISP (Hispanic origin, 01 = not Hispanic), PWGTP (person weight — used to compute population-representative proportions). Hispanic origin is given priority over RAC1P per Census Bureau convention, producing a unified race/ethnicity label. These weights connect the PUMS data to NHANES in the joined EDA, satisfying the Multiple Datasets challenge goal.

Race/Ethnicity	WA Proportion	National Proportion	Difference
White	63.3%	57.3%	+6.0 pp
Hispanic	14.0%	18.9%	-4.9 pp
Asian	10.3%	6.1%	+4.2 pp
Two or More Races	6.9%	4.3%	+2.6 pp
Black or African American	3.5%	12.3%	-8.8 pp
Native Hawaiian/Pacific Islander	0.7%	0.2%	+0.5 pp
Native American	0.6%	0.4%	+0.2 pp
Other	0.6%	0.5%	+0.1 pp
Alaska Native	0.1%	0.0%	+0.1 pp

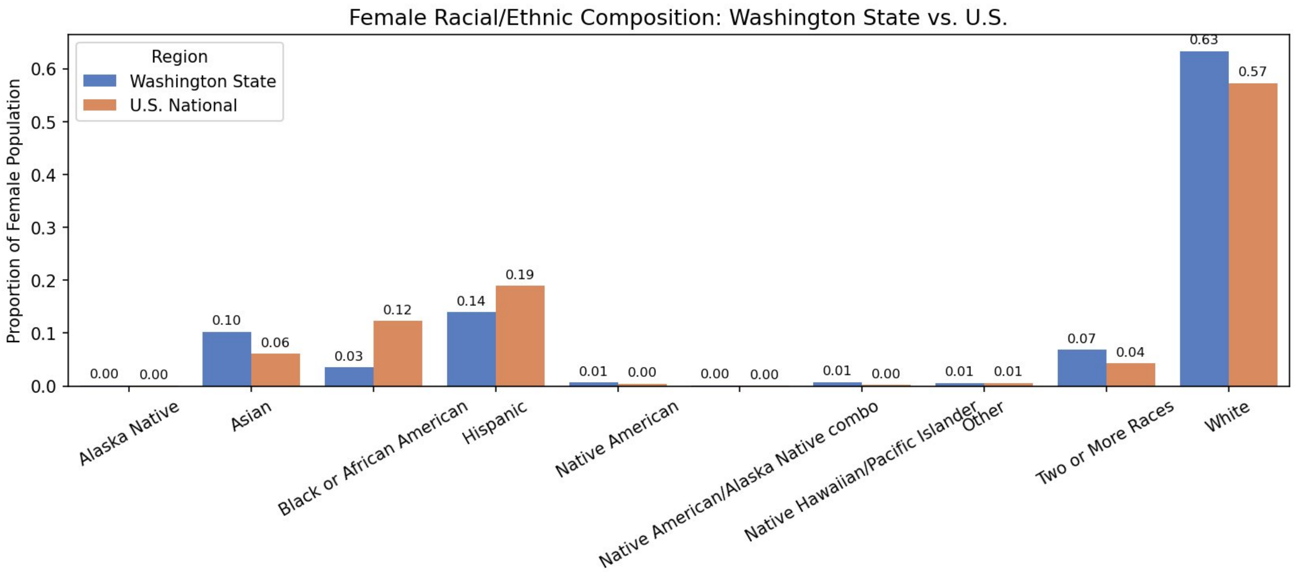

__Figure 5.__ Female racial/ethnic composition of Washington state vs. the U.S., computed from ACS PUMS 2020-2024 using PWGTP person weights. Washington state has a notably higher proportion of Asian women (10.3% vs. 6.1%) and a substantially lower proportion of Black or African American women (3.5% vs. 12.3%) and Hispanic women (14.0% vs. 18.9%) compared to the national average.

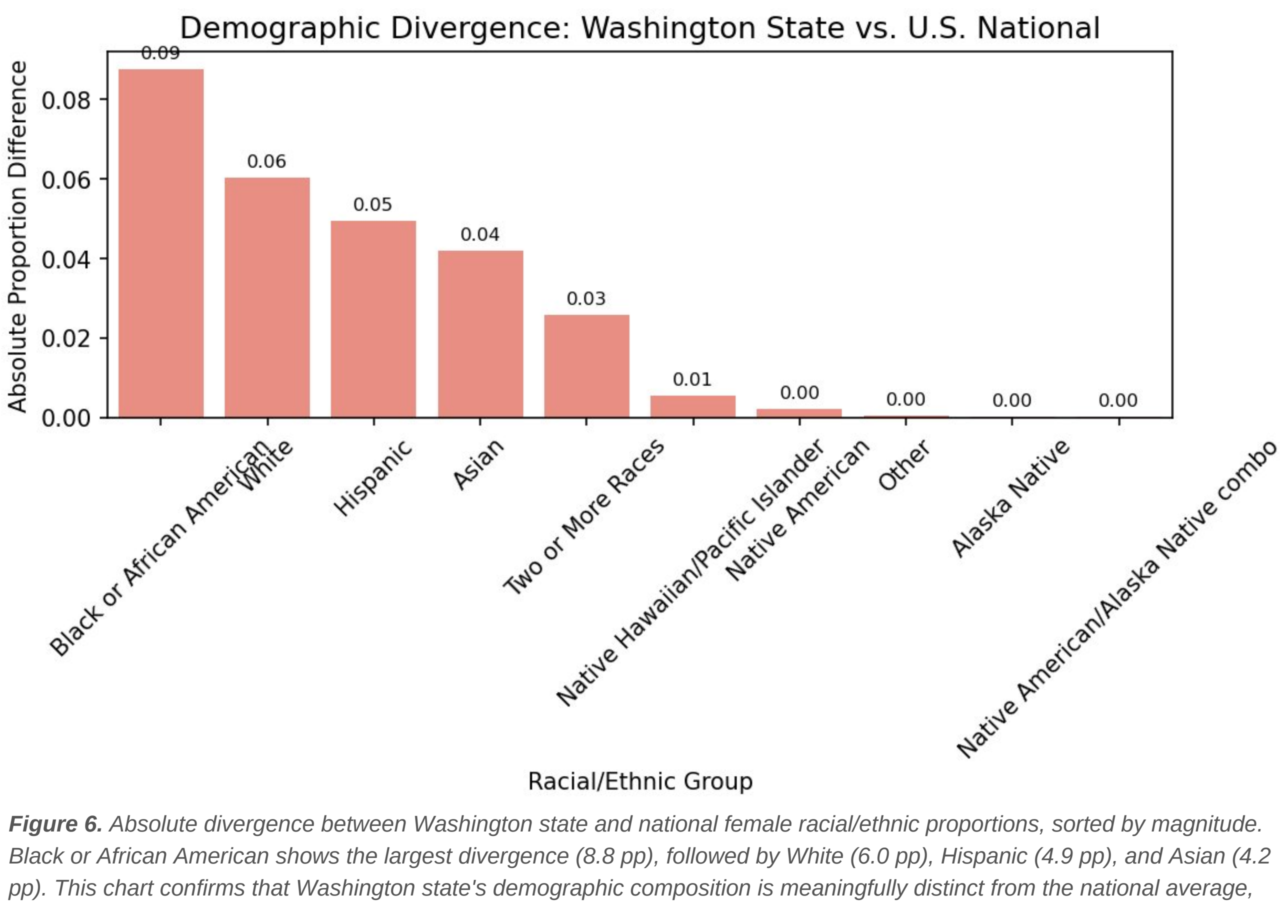

__Figure 6.__ Absolute divergence between Washington state and national female racial/ethnic proportions, sorted by magnitude. Black or African American shows the largest divergence (8.8 pp), followed by White (6.0 pp), Hispanic (4.9 pp), and Asian (4.2 pp). This chart confirms that Washington state's demographic composition is meaningfully distinct from the national average, which forms the basis for the Washington-weighted body measurement estimates in the joined EDA.

5.4 Joined Datasets: NHANES + ACS PUMS

Dataset relationship: NHANES provides individual-level body measurements grouped by race/ethnicity (RIDRETH3). ACS PUMS provides the racial/ethnic composition of Washington state's and the U.S.'s female populations via PWGTP person weights. The two datasets are joined conceptually by mapping NHANES RIDRETH3 codes to PUMS race/ethnicity labels, then applying PUMS-derived proportions as weights to NHANES group means — producing a Washington state-representative body measurement estimate. Note: ANSUR II is excluded from the joined EDA because it contains no race/ethnicity variable; its structural measurements (sleeve, inseam, shoulder width) cannot be demographically weighted.

Measurement	WA Weighted Mean	U.S. National Mean	Difference
BMXWAIST (waist, mm)	983.2	993.1	-9.9 mm
BMXHT (height, mm)	1601.7	1602.6	-0.9 mm
BMXWT (weight, kg)	75.9	77.4	-1.5 kg

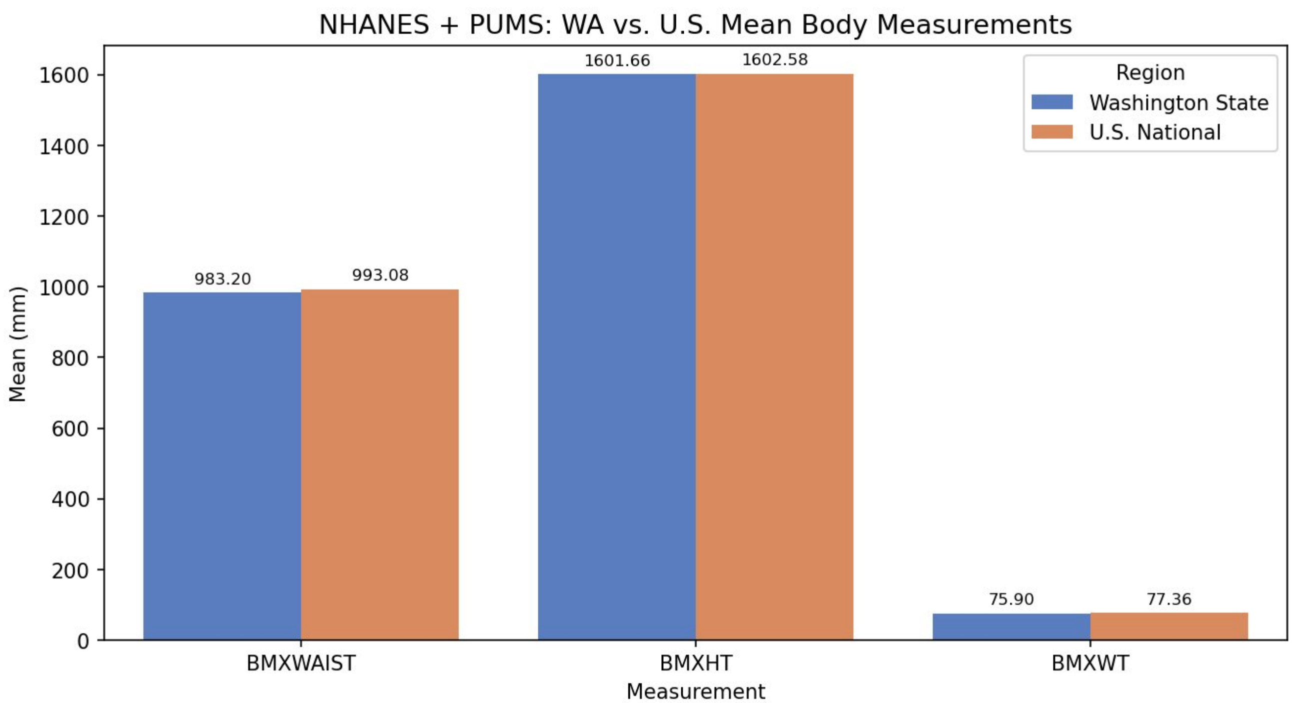

Figure 7. *Washington state-weighted vs. U.S. national mean body measurements estimated by joining NHANES 2015-2018 with ACS PUMS 2020-2024 demographic weights. Washington state women show a smaller estimated waist circumference (983.2 vs. 993.1 mm, a difference of 9.9 mm) and lower weight (75.9 vs. 77.4 kg), while height is nearly identical. The waist difference is the most relevant finding for retail sizing implications and will be tested for statistical significance in RQ4.*

5.5 Ann Taylor Size Guide

Size: Four DataFrames manually transcribed from the Ann Taylor size chart modal (accessed via any product page at anntaylor.com, May 2026): classic_tops (11 rows x 8 columns), classic_bottoms (11 rows x 10 columns), petite_tops (10 rows x 8 columns), petite_bottoms (11 rows x 10 columns). Each row represents one numeric size. Columns include size labels, original inch measurements, and converted cm measurements.

Missing data: No missing values in any of the four DataFrames — all values were manually transcribed from the published size guide and verified against the Ann Taylor website.

Variables of interest: bust_cm, sleeve_cm, waist_cm (tops); waist_regular_cm, waist_curvy_cm, hip_regular_cm, hip_curvy_cm (bottoms). These are compared against Washington state women's estimated measurements in RQ2. The Curvy fit subcategory — discovered during EDA — offers a smaller waist with larger hip allowance and is particularly relevant to petite body proportions.

Variable	Mean (cm)	Std	Min	Median	Max
Classic bust_cm	94.2	10.6	80.0	92.7	113.0
Classic sleeve_cm	79.0	2.5	75.6	78.7	83.2
Classic waist_cm	76.4	10.6	62.2	74.9	95.3
Petite bust_cm	89.8	9.0	77.5	88.9	105.4
Petite sleeve_cm	74.8	2.2	71.8	74.6	78.4
Petite waist_cm	72.3	8.8	59.7	71.1	87.6

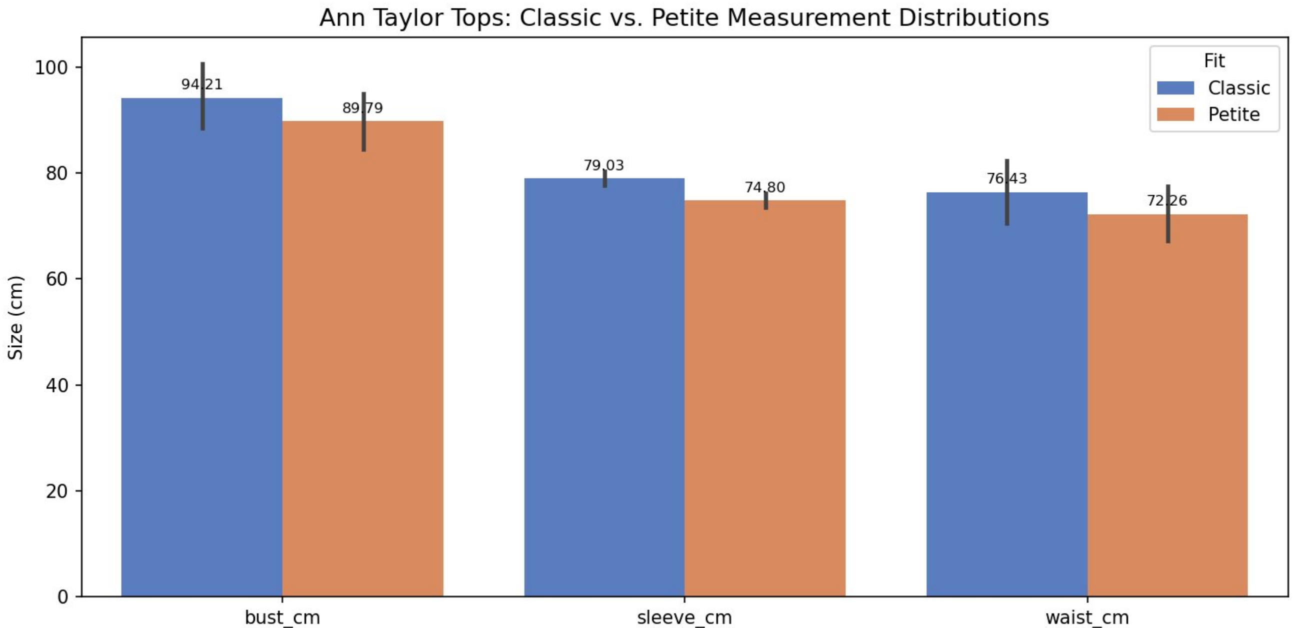

__Figure 8.__ Mean Ann Taylor Classic vs. Petite top measurements (cm) across all sizes. The Petite line is consistently smaller across bust (89.8 vs. 94.2 cm), sleeve (74.8 vs. 79.0 cm), and waist (72.3 vs. 76.4 cm). The sleeve difference (~4.2 cm) is particularly notable, as it directly reflects the shorter arm lengths of petite women.

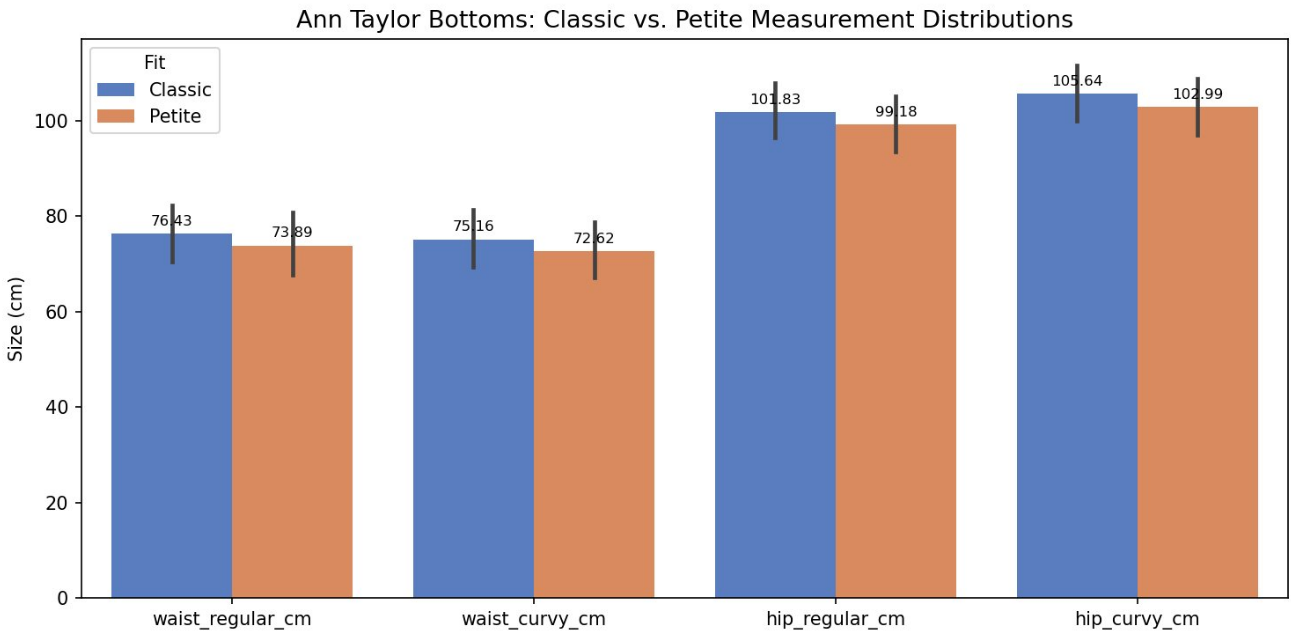

Figure 9. *Mean Ann Taylor Classic vs. Petite bottoms measurements (cm) for both Regular and Curvy fits. Petite measurements are consistently smaller across all four dimensions. The Curvy fit shows a smaller waist with a proportionally larger hip measurement compared to Regular within the same size, accommodating a different waist-to-hip ratio.*

5.6 Result Validity: Null Hypotheses and Test Assumptions

For RQ4, independent samples t-tests will be applied. The null and alternative hypotheses for each measurement are:

Measurement	Null Hypothesis (H0)	Alternative (H1)
Waist (BMXWAIST)	Mean WA waist = mean U.S. waist	Mean WA waist < mean U.S. waist
Height (BMXHT)	Mean WA height = mean U.S. height	Mean WA height < mean U.S. height
Weight (BMXWT)	Mean WA weight = mean U.S. weight	Mean WA weight < mean U.S. weight

Test: scipy.stats.ttest_ind(), significance level alpha = 0.05. Cohen's d effect sizes will be computed alongside p-values to confirm practical as well as statistical significance.

Assumptions verified during EDA: (1) **Independence** — NHANES uses a complex survey design; individual rows are treated as independent observations for this analysis, which is a known simplification. (2) **Approximate normality** — Figure 3 shows BMXHT is approximately normally distributed; BMXWAIST is right-skewed, but the large sample size (n > 700 per group) justifies the t-test via the Central Limit Theorem. (3) **Equal variances** — Levene's test will be conducted in the final deliverable to determine whether to use Student's or Welch's t-test.

6. Challenge Goals Update

1. Multiple Datasets

On track. Three datasets (ANSUR II, NHANES, ACS PUMS) are loaded and used in the EDA, and the Ann Taylor size guide adds a fourth reference source. The explicit join between NHANES and PUMS is implemented in run_joined_eda(), which applies PUMS demographic weights to NHANES group means. One key limitation discovered during EDA: ANSUR II contains no race/ethnicity column, so ANSUR II structural measurements cannot be demographically weighted. This will be clearly acknowledged as a limitation in the final deliverable.

2. Result Validity

On track. Null hypotheses are stated above (Section 5.6). t-test assumptions have been partially verified through distribution plots (Figures 3-4). Levene's test for equal variances and full t-test implementation with Cohen's d will be completed in the final deliverable.

7. Work Plan Evaluation

Task 1 (data acquisition and setup) took approximately 3 hours as estimated. An unexpected challenge was the national PUMS file being split into four parts (psam_pusa-d.csv), requiring concatenation logic not anticipated in the original plan. Task 2 (cleaning and preprocessing) ran longer than estimated (~7 hours vs. 5) due to the discovery that ANSUR II lacks a race/ethnicity variable, requiring a significant restructuring of the joined EDA approach. The Ann Taylor size guide also expanded from one DataFrame to four after discovering the Curvy fit subcategory. Tasks 3-5 are partially complete through the EDA phase. Remaining work includes the full RQ2 size alignment analysis, RQ4 statistical testing, and the final report.

Task	Original Est.	Actual / Remaining	Status
1. Data acquisition + setup	3 hrs	3 hrs	Complete
2. Cleaning + preprocessing	5 hrs	7 hrs	Complete
3. RQ1 + RQ3 analysis	6 hrs	6 hrs (EDA done)	In progress
4. RQ2 Ann Taylor alignment	4 hrs	4 hrs remaining	Not started
5. RQ4 statistical testing	5 hrs	5 hrs remaining	Not started
6. Report + code cleanup	4 hrs	5 hrs remaining	In progress

8. Testing

A separate test_eda.py module contains 6 unit tests verifying the core analysis functions in analysis.py. All tests use small synthetic DataFrames with known expected outputs, allowing fast and deterministic verification independent of the full datasets. All 6 tests pass.

test_summarize_dataset: Verifies seven-number summary values (mean height = 165.0, mean waist = 67.5) and categorical counts (White count = 2) on a 4-row synthetic DataFrame.

test_compute_demographic_weights: Verifies weights sum to 1.0, Hispanic and Asian labels appear, and Hispanic proportion = 150/500 = 0.30 on a known input.

test_compute_weighted_means: Verifies weighted mean with equal 50/50 group weights: expected waist = 700.0 mm and height = 1600.0 mm.

test_load_filters_females_only: Simulates SEX == 2 filter and verifies row count = 3 and all rows are female.

test_missing_data_detection: Verifies isnull().sum() correctly identifies 1 missing value per column and 2 total.

test_load_anntaylor: Verifies all four DataFrames have correct row counts (11, 11, 10, 11) and that inch-to-cm conversion is correct: 24.5 in x 2.54 = 62.23 cm.

9. Collaboration

Claude (Anthropic) was used as an AI assistant throughout this project to help structure research questions, draft report sections, review Python code for correctness and CSE 163 style compliance, and explain unfamiliar functions and concepts (e.g., melt(), demographic weighting, t-test assumptions). All datasets were independently downloaded from their respective public sources. All code was written, reviewed, and understood by the author before submission. No other people or outside resources were consulted.